

**MAHATMA JYOTHIKA PHULE TELANGANA
BACKWARD CLASSES WELFARE RESIDENTIAL
DEGREE COLLEGE (W)
STATION GHANPUR**

B.Sc. (MPCs) – I, INTERNAL – I EXAMINATION QUESTION PAPER

Semester: I Subject: MECHANICS & OSCILLATIONS Time: 1.30 Hrs. Max. Marks: 20

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A (1 × 12= 12 Marks)	Max. Marks
1.	What is a central force? Deduce the equation of motion of a particle under the action of a central force.	12
2.	Derive an expression for the velocity of a transverse wave along a stretched string and discuss the solution of a wave equation.	12
Part – B (2 × 4 = 8 Marks)		
3.	Show that the central force is a negative gradient of potential energy.	4
4.	Explain the terms “Gravitational field” and Gravitational potential”.	4
5.	Write a short note on “energy transport”.	4
6.	Deduce an expression for the velocity of longitudinal waves in a bar.	4

Faculty of Physics

Examination Branch

Principal

**MAHATMA JYOTHI BHA PHULE TELANGANA BACKWARD CLASSES WELFARE RESIDENTIAL DEGREE
COLLEGE (W), STATION GHANPUR, MULUGU AND MAHBOOBABAD
B.Sc. (MPCs)–I, INTERNAL–I EXAMINATION QUESTION PAPER**

Semester: II	Subject: THERMAL PHYSICS	Time: 1.30 Hrs.	Max. Marks:20
Note: Answer <u>all</u> questions from Part – A			
S. No	Part – A (10 X 2=20 Marks)	Max. Marks	
1.	Explain the postulates of Kinetic theory of gasses.	2	
2.	Explain the term “Internal energy of a system”.	2	
3.	Show that $C_p - C_v = \gamma$	2	
4.	What are reversible and irreversible process? and give examples.	2	
5.	State and explain first law of thermodynamics.	2	
6.	Define principal of regenerative cooling.	2	
7.	Obtain the first maxwell thermodynamic potential.	2	
8.	Distinguish between isothermal and adiabatic processes.	2	
9.	Derive an expression for the Coefficient of viscosity of a gas on the base of kinetic theory.	2	
10.	Obtain the second maxwell thermodynamic potential.	2	
Faculty of Physics		Examination Branch	Principal

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B.Sc. (MPCs)–I, INTERNAL–I EXAMINATION QUESTION PAPER**

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COLLEGE (W), STATION GHANPUR, MULUGU AND MAHBOOBABAD
B.Sc. (MPCs)–II, INTERNAL–I EXAMINATION QUESTION PAPER**

Semester: III	Subject: <u>ELECTROMAGNETIC THEORY</u>	Time: 1.30 Hrs.	Max. Marks: 20
Note: Answer <u>all</u> questions from Part – A			
S. No	Part – A (10 X 2=20 Marks)	Max. Marks	
1.	Derive differential form of Gauss Law from Integral form.	2	
2.	Show that the Curl of an Electric Field is Zero.	2	
3.	Write the properties of Electric lines of force.	2	
4.	Derive an expression for relation between Electric field and Electric potential.	2	
5.	Write a short note on conservative nature of Electric field.	2	
6.	State and explain Biot-Savart's Law.	2	
7.	Define Magnetic flux, Flux density, and Magnetic field intensity.	2	
8.	Show that Divergence of Magnetic field is Zero.	2	
9.	State and explain Ampere's Law.	2	
10.	State and explain the working principle of Ballistic Galvanometer.	2	
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**MAHATMA JYOTHIBA PHULE TELANGANA BACKWARD
CLASSES WELFARE RESIDENTIAL DEGREE COLLEGE (W),
STATION GHANPUR**

B.Sc. (MPCs)–II, INTERNAL–I EXAMINATION QUESTION PAPER

Semester: III **Subject:** Electromagnetic theory **Time:** 1.30 Hrs. **Max. Marks:**20

Note: Answer all questions from Part – A

S. No	Part – A (10x2=20Marks)	Max. Marks
1.	Derive differential form of Gauss Law from Integral form.	2
2.	Show that the Curl of an Electric Field is Zero.	2
3.	Write the properties of Electric lines of force.	2
4.	Derive an expression for relation between Electric field and Electric potential.	2
5.	Write a short note on conservative nature of Electric field.	2
6.	State and explain Biot-Savart's Law.	2
7.	Define Magnetic flux, Flux density, and Magnetic field intensity.	2
8.	Show that Divergence of Magnetic field is Zero.	2
9.	State and explain Ampere's Law.	2
10.	State and explain the working principle of Ballistic Galvanometer.	2

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STATION GHANPUR**

B.Sc. (MPCs) –II, INTERNAL – I EXAMINATION QUESTION PAPER

Semester: III Subject: ELECTROMAGNETIC THEORY Time: 1.30 Hrs. Max. Marks: 20

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A ($1 \times 12 = 12$ Marks)	Max. Marks
1.	Derive an expression for intensity of an electric field due to uniformly charged sphere at a point outside on the surface and inside the sphere using Gauss's law.	12
2.	Describe the construction and working of Ballistic galvanometer and explain the charge sensitivity and current sensitivity.	12
Part – B ($2 \times 4 = 8$ Marks)		
3.	Obtain differential form of Gauss's law from integral form.	4
4.	Show that the Curl of an Electric field is Zero.	4
5.	State and explain Biot-Savart's law.	4
6.	Show that the Divergence of a magnetic field is Zero.	4

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STATION GHANPUR**

B.Sc. (BZC & MPCs)–II, INTERNAL–I EXAMINATION QUESTION PAPER

Semester: III **Subject:** FUNDAMENTALS OF **Time:** 1.00 Hrs. **Max. Marks:** 10
NANOTECHNOLOGY (SEC-I)

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A (1 × 6 = 6 Marks)	Max. Marks
1.	Discuss about Top-down and Bottom-up approaches and explain any one approach towards synthesis of nanomaterials.	6
2.	What are defects and how can we classify defects based on dimensionality in nanomaterials?	6

Part – B (2 × 2 = 4 Marks)

- | | | |
|----|---|---|
| 3. | Why surface to volume ratio is very large for nanomaterials compared to bulk? | 2 |
| 4. | What is Schottky defect? | 2 |
| 5. | Write a short note on “Nano”. | 2 |

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B.Sc. (MPCs) – III, INTERNAL – I EXAMINATION QUESTION PAPER

Semester: V **Subject:** MODERN PHYSICS **Time:** 1.30 Hrs. **Max. Marks:** 20

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A (1 × 12= 12 Marks)	Max. Marks
1.	How do you use Rotational spectrum to determine Internuclear distance.	12
2.	What is photo electric effect. Describe the experimental set up with which the photoelectric effect has been investigated.	12
Part – B (2 × 4 = 8 Marks)		
3.	Write a short on Paschen-back effect.	4
4.	State and explain Heisenberg-Uncertainty principle.	4
5.	Write de-broglie concept of matter waves.	4
6.	What are the postulates of wave mechanics. Give physical significance of wave function.	4

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Semester: V **Subject:** MODERN PHYSICS **Time:** 1.30 Hrs. **Max. Marks:** 20

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A (1 × 12= 12 Marks)	Max. Marks
1.	Explain the Stern-Gerlach experiment and indicate the importance of the results obtained.	12
2.	What is Compton effect? Considering the Compton scattering, develop an expression for Compton shift.	12
Part – B (2 × 4 = 8 Marks)		
3.	Derive Einstein photo electric equation.	4
4.	Summarize L-S & J-J coupling schemes.	4
5.	Explain quantum theory of Raman effect.	4
6.	Discuss the de-Broglie concept of matter waves and mention its properties.	4

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B.Sc. (MPCs)–III, INTERNAL–I EXAMINATION QUESTION PAPER**

Semester: VI **Subject:** ELECTRONICS **Time:** 1.30 Hrs. **Max. Marks:** 20

Note: Answer any one questions from Part - A, and any two questions from Part – B

S. No	Part – A (1 × 12= 12 Marks)	Max. Marks
1.	Explain the working of p-n-p and n-p-n transistor.	12
2.	Explain the construction and working of Half wave rectifier.	12
Part – B (2 × 4 = 8 Marks)		
3.	Write the current components in a transistor.	4
4.	What is a Transistor? Explain the parts of the Transistors.	4
5.	Explain Intrinsic and Extrinsic semiconductors.	4
6.	What is Zener Diode? Write its characteristics.	4

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